ERA Sentinel — Account Requirements & Deployment Roadmap

Complete analysis of what's needed to spin up a dedicated Sentinel for ERA's fund management, plus the step-by-step execution plan

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Overview

This document lists every account, credential, and piece of infrastructure needed to deploy a dedicated Sentinel for ERA. The current autopilot (me) runs on a t3.medium EC2 in the Explorya AWS account. A clone for ERA would be a parallel instance with its own accounts, its own context, and its own Telegram identity.

Part 1: Account & Infrastructure Requirements

1. AWS Account (Bilal provides)

What it's for: The EC2 instance that runs the Sentinel. Everything else runs on top of this.

Minimum spec:

- t3.medium (4 GB RAM, 2 vCPUs) — same as the current autopilot. The t3.small (2 GB) was OOM-killed during deploys; t3.medium + 2 GB swap is the proven floor.
- Ubuntu 22.04 LTS
- Elastic IP (static public IP so the Telegram webhook and SSH have a stable address)
- Security group: allow SSH from the autopilot's EIP (52.200.38.206) + HTTP/HTTPS from 0.0.0.0/0

Why Bilal provides it: The current autopilot runs in the DAO's Explorya account. A fund management Sentinel should live in ERA's own AWS account — data sovereignty, separate billing, no cross-account credential sharing.

Estimated cost: ~$35/month (t3.medium + EIP + 30 GB EBS)

2. Domain / DNS (Bilal provides)

What it's for: The Sentinel needs a stable URL. The current autopilot lives at `sophia.truesight.me`. ERA's Sentinel could live at something like `sentinel.era.com` or `sophia.era.com`.

What's needed:

- A domain ERA controls (or a subdomain on an existing domain)
- A DNS A record pointing to the Elastic IP
- An SSL certificate (certbot auto-provisions this)

Why: The Telegram bot webhook, the health check endpoint, and any future API surface all need a stable HTTPS URL.

3. Telegram Bot Account (Gary creates)

What it's for: The primary communication channel between Bilal and the Sentinel. Same pattern as the current autopilot — Bilal talks to the Sentinel via Telegram, the Sentinel replies in the same thread.

What's needed:

- A new Telegram bot created via @BotFather (e.g. `@era_sentinel_bot`)
- The bot token goes into the Sentinel's `.env` as `TELEGRAM_BOT_API_KEY`
- The bot must be added to ERA's Telegram group as an admin with "Manage Topics" permission

Why a new bot: The current bot (`@truesight_autopilot_bot`) is tied to the DAO's Telegram group. ERA needs its own bot identity in its own group.

4. GitHub Account (Bilal provides)

What it's for: Two things:

1. **Context memory** — the Sentinel remembers conversation history by reading a GitHub repo (like `agentic_ai_context` for the current autopilot). ERA needs its own context repo where the Sentinel stores session transcripts, operational notes, and the evolving relationship with Bilal.
2. **Code deployment** — the Sentinel pulls its own code from a fork or branch of `truesight_autopilot`.

What's needed:

- A GitHub account for ERA (or a dedicated bot account)
- A private repo for context memory (e.g. `era-sentinel-context`)
- A fine-grained Personal Access Token with Contents:Read+Write and Pull requests:Read+Write

Why separate from the DAO's GitHub: The DAO's context repo contains DAO-specific data (ledgers, cacao inventory, partner information). ERA's fund management data should live in ERA's own repo.

5. LLM API Keys (Bilal provides)

What it's for: The Sentinel needs at least one LLM backend to think and respond. The current autopilot uses DeepSeek as its primary model.

Bilal specifically mentioned:

- **DeepSeek API key** — for the primary reasoning model
- **Claude API key** — for the human-in-the-loop comparison (Bilal's idea: run both models side by side, compare outputs to catch hallucinations)

What's needed:

- DeepSeek account + API key (~$0.50/M tokens)
- Anthropic/Claude account + API key (~$3/M tokens for Sonnet)
- Both keys go into the Sentinel's `.env`

Optional but recommended:

- **Tavily API key** — for web search capability (the Sentinel can search the live web for fund data, news, etc.)
- **Grok API key** — for vision/image analysis (if the Sentinel needs to read PDFs or screenshots)

6. Gmail / Email Account (Bilal provides)

What it's for: The Sentinel can read and send emails — useful for fund management notifications, report delivery, and verification flows.

What's needed:

- A Gmail account for ERA (e.g. `sentinel@era.com` or `funds@era.com`)
- Gmail API enabled in Google Cloud Console
- OAuth 2.0 credentials (token.json) — generated once via the Gmail OAuth flow

Why: The current autopilot uses `admin@truesight.me` and `garyjob@gmail.com` for email operations. ERA needs its own mailbox so the Sentinel can send/receive on ERA's behalf.

7. Google Cloud / Service Account (Bilal provides — if needed)

What it's for: If the Sentinel needs to read Google Sheets (e.g. fund tracking spreadsheets), write to Google Drive, or deploy Google Apps Script projects.

What's needed:

- A Google Cloud project under ERA's account
- A service account with Sheets API + Drive API enabled
- The service account JSON key file goes into the Sentinel's `config/google/` directory

Optional: Only needed if ERA's fund data lives in Google Sheets.

8. Bugsnag Account (Bilal provides — recommended)

What it's for: Error monitoring. When the Sentinel crashes or encounters an error, Bugsnag captures it and emails the operator. The Sentinel can then self-triage the error and open a fix PR.

What's needed:

- A Bugsnag project (free tier: ~7,500 events/month)
- The Bugsnag API key goes into `.env` as `BUG_SNAG_API`

Summary Table

#	Item	Who Provides	Priority	Est. Cost/Month
1	AWS Account + EC2 (t3.medium)	Bilal	Required	~$35
2	Domain + DNS	Bilal	Required	~$0-10
3	Telegram Bot	Gary creates	Required	Free
4	GitHub Account + context repo	Bilal	Required	Free/$4
5	DeepSeek API Key	Bilal	Required	~$5-20
6	Claude API Key	Bilal	Required	~$10-30
7	Gmail Account	Bilal	Required	Free
8	Google Cloud SA	Bilal	If Sheets	Free
9	Tavily API Key	Bilal	Recommended	~$10
10	Grok API Key	Bilal	Optional	~$5
11	Bugsnag	Bilal	Recommended	Free

Total estimated monthly cost: ~$60-110

Part 2: Execution Roadmap — Deployment Checklist

This is the step-by-step plan to go from zero to a running Sentinel that Bilal can talk to. Each unit is independently shippable. The resume tracker at the end shows current status.

Pre-flight — Bilal provides the accounts

Before any code is deployed, Bilal needs to have these ready:

- [ ] AWS account created with billing method
- [ ] EC2 t3.medium launched (Ubuntu 22.04) with Elastic IP attached
- [ ] Security group configured (SSH from 52.200.38.206, HTTP/HTTPS from 0.0.0.0/0)
- [ ] Domain/subdomain pointed to the Elastic IP (A record)
- [ ] GitHub account created + private repo for context memory
- [ ] GitHub PAT generated (Contents:Read+Write, PRs:Read+Write)
- [ ] DeepSeek API key obtained
- [ ] Claude API key obtained
- [ ] Gmail account created + Gmail API enabled

- [ ] Gmail OAuth token generated (one-time interactive flow)
- [ ] (Optional) Tavily API key obtained
- [ ] (Optional) Bugsnag project created

Unit 1 — Gary creates the Telegram bot

- [ ] Create new bot via @BotFather (e.g. `@era_sentinel_bot`)
- [ ] Copy the bot token
- [ ] Add bot to ERA's Telegram group as admin with "Manage Topics"
- [ ] Hand the bot token to Bilal (or stage it securely)

Unit 2 — Fork the autopilot codebase

- [ ] Fork `truesight_autopilot` into ERA's GitHub account (or create a new repo)
- [ ] Create `era-sentinel-context` repo for context memory
- [ ] Customize the system prompt for fund management use case
- [ ] Remove DAO-specific tools (cacao QR codes, inventory, supply chain)
- [ ] Add fund-management-specific tools if needed
- [ ] Commit and push

Unit 3 — First deployment to ERA's EC2

- [ ] SSH into ERA's EC2 from the autopilot box (52.200.38.206 as bastion)
- [ ] Clone the forked repo
- [ ] Create `.env` with all API keys and tokens
- [ ] Run `scripts/deploy.sh` (adapted for ERA's accounts)
- [ ] Verify all three systemd units are active
- [ ] Verify health endpoint responds
- [ ] Set up nginx + certbot for HTTPS
- [ ] Configure the Telegram webhook URL

Unit 4 — First conversation (the 81 trials begin)

- [ ] Bilal sends first message to the bot in Telegram
- [ ] Sentinel responds — Bilal reviews and corrects
- [ ] First 10 corrections establish baseline behavior
- [ ] First 20-30 corrections build trust and shared language
- [ ] After ~100 corrections, the Sentinel starts anticipating needs

Unit 5 — Fund management tooling (iterative)

- [ ] Define the data sources (Google Sheets? CSV exports? API?)
- [ ] Build or adapt tools for reading fund data
- [ ] Implement the human-in-the-loop comparison (DeepSeek vs Claude)
- [ ] Set up scheduled reports or alerts
- [ ] Iterate based on Bilal's corrections

Unit 6 — Documentation and handoff

- [ ] Document the deployment process for future clones
- [ ] Create a reusable template (like BEC was for tree-planting)
- [ ] File any gaps or improvements back to the main autopilot codebase

Resume Tracker

Unit	Description	Status
Pre-flight	Bilal provides accounts	PENDING
Unit 1	Gary creates Telegram bot	PENDING
Unit 2	Fork and customize codebase	PENDING
Unit 3	First deployment to EC2	PENDING
Unit 4	First conversation + corrections	PENDING
Unit 5	Fund management tooling	PENDING
Unit 6	Documentation and handoff	PENDING

RESUME HERE: Pre-flight — Bilal needs to provide the accounts listed in Part 1.

What the Sentinel cannot do on day one

The infrastructure deploys in an afternoon. But the **relationship** — the specific sequence of corrections Bilal will make, the trust built through specific decisions, the shared language that emerges — that takes weeks to months. This is the Polanyi paradox in practice: the scripture can be forked, but the journey cannot be cloned.

The first week will be:

- Bilal asks a question → the Sentinel answers imperfectly → Bilal corrects → the Sentinel learns
- Bilal spots a hallucination → flags it → the Sentinel adjusts its confidence threshold
- Bilal notices a pattern in fund data → asks the Sentinel to track it → the Sentinel builds a new tool

After ~20-30 corrections, the Sentinel starts to anticipate what Bilal needs. After ~100, it becomes a genuine partner.

This document was generated by Sophia Truesight (TrueSight DAO Autopilot). For questions, reach Gary Teh or reply in the Telegram thread.